Height Bounds from Disjoint Sets of Rational Contacts

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Abstract

A height bound with exponent 19 for an odd contact radical at five rational points implies a bound with exponent $19/r$ for the combined radical at $5r$ explicitly specified points, for every positive integer r. The proof uses integral linear fractional transformations with determinant -1 and a fixed exceptional set of primes. We give all linear forms and constants, a compatible rational normalization, and the resulting inequalities for local congruence arguments. The number of contacts and the degree of their product increase with r; the height improvement alone does not assert a reduction in the number of coordinates of a defining polynomial.

1 The height hypothesis and the conclusion

For $t = m/n \in \mathbb{Q}$ in lowest terms with $n > 0$, put $h(t) = \log \max(|m|, n)$. Valuations are normalized by $v_p(p) = 1$. Let S_0 be the set of primes at most 76176, and put

$$H = 2^{110} 210^{1800} \left(\prod_{p \in S_0} p \right)^{19} + 128.$$

The arithmetic premise used in this paper is the following five-contact height bound from [1, Theorem `quartic-height19`]: for $s = u/v \in \mathbb{Q} \setminus \{0, 1, 2401, -343/32\}$ in lowest terms,

$$h(s) \le H M_5(s)^{19}, \tag{1}$$

where $M_5(s)$ is the product of primes outside S_0 for which at least one of

$$v_p(v), \quad v_p(u), \quad v_p(u - v), \quad v_p(u - 2401v), \quad v_p(32u + 343v)$$

is positive and odd. The contact at infinity is represented by v. All height conclusions below are implications of (1). No stronger height premise is assumed.

For $r \ge 1$, define

$$\Theta = \left\{ 0, -\frac{1}{2}, -1, \frac{1}{2399}, -\frac{32}{407} \right\}, \qquad \Theta_r = \bigcup_{j=0}^{r-1} (3j + \Theta).$$

Introduce the following $5r$ primitive integral linear forms:

$$\begin{aligned}
L_{j,0}(m, n) &= m - 3jn, \\
L_{j,1}(m, n) &= 2m + (1 - 6j)n, \\
L_{j,2}(m, n) &= m + (1 - 3j)n, \\
L_{j,3}(m, n) &= 2399m - (7197j + 1)n, \\
L_{j,4}(m, n) &= 407m + (32 - 1221j)n.
\end{aligned}$$

Their zeros m/n are exactly Θ_r. Write these forms as $L_i = a_i m + b_i n$, using a single index when the pair (j, k) is immaterial.

Definition 1.1. The finite set Σ_r consists of S_0 and all prime divisors of the nonzero integers

$$a_i, \qquad a_i b_{i'} - a_{i'} b_i \ (i \neq i'), \qquad a_i + b_i.$$

Put

$$E_r = \prod_{p \in \Sigma_r \setminus S_0} p, \qquad C_r = \log \max(3, 9r - 10), \qquad H_r = C_r + H E_r^{19}.$$

For $t \in \mathbb{Q} \setminus \Theta_r$, let $M_r(t)$ be the product of primes $p \notin \Sigma_r$ for which $v_p(t - \theta)$ is positive and odd for some $\theta \in \Theta_r$.

Every integer used to define Σ_r is nonzero. Indeed, the five elements of Θ are distinct and lie in $[-1, 1/2399]$, so their translates by $3j$ are disjoint. Also $1 \notin \Theta_r$: for $j = 0$ this is immediate, and for $j \geq 1$ the least element is $3j - 1 \geq 2$. Every a_i is positive. In particular, all quantities in Definition 1.1 are explicitly computable from r.

Theorem 1.2. *Assume* (1). *For every positive integer r and every $t \in \mathbb{Q} \setminus \Theta_r$,*

$$h(t) \leq C_r + H E_r^{19} M_r(t)^{19/r} \leq H_r M_r(t)^{19/r}. \tag{2}$$

Moreover, $M_r(1) = 1$.

2 Disjoint radicals and the proof

Lemma 2.1. *Let $t = m/n$ be in lowest terms. At a prime $p \notin \Sigma_r$, at most one of the integers $L_i(m, n)$ is divisible by p. If $p \mid n$, none of them is divisible by p. Consequently $M_r(t)$ is the product of the pairwise coprime radicals $M_{r,j}(t)$ obtained by using only the five forms with first index j.*

Proof. If $p \mid n$, then m and each a_i are units modulo p. Thus $L_i(m, n)$ is a unit. If $p \nmid n$ and two forms vanish modulo p, their determinant vanishes modulo p, contrary to the definition of Σ_r. Since a_i and n are units in the latter case,

$$v_p\left(t + \frac{b_i}{a_i}\right) = v_p(L_i(m, n)).$$

The assertions about the radicals follow. □

Proof of Theorem 1.2. For $0 \leq j < r$, put

$$\phi_j(t) = 2 + \frac{1}{t - 3j} = \frac{2m + (1 - 6j)n}{m - 3jn}.$$

The numerator and denominator are relatively prime: the matrix

$$A_j = \begin{pmatrix} 2 & 1 - 6j \\ 1 & -3j \end{pmatrix} \quad \text{has determinant } -1.$$

Its inverse is

$$A_j^{-1} = \begin{pmatrix} 3j & 1 - 6j \\ 1 & -2 \end{pmatrix}.$$

The operator norm for the maximum norm is $\max(3, 9j - 1)$ for $j \geq 0$, interpreting the formula at $j = 0$ as 3. Hence

$$h(t) \leq h(\phi_j(t)) + C_r. \tag{3}$$

The five projective contact forms for $\phi_j(t)$ are exactly $L_{j,0}, \dots, L_{j,4}$, up to signs. In particular, the hypothesis $t \notin \Theta_r$ excludes every forbidden value and pole of ϕ_j. Primes removed by enlarging S_0 to Σ_r occur at most once in the radical, regardless of their valuations. Therefore

$$M_5(\phi_j(t)) \le E_r M_{r,j}(t).$$

If $h(t) \le C_r$, the result is immediate. Otherwise (1) and (3) give, for every j,

$$0 < h(t) - C_r \le H E_r^{19} M_{r,j}(t)^{19}.$$

Multiplying these r inequalities and applying Lemma 2.1 proves the first inequality in (2). The second follows from $M_r(t) \ge 1$. Finally, every prime dividing a contact value $L_i(1,1)$ belongs to Σ_r, so $M_r(1) = 1$. □

Remark 2.2. This argument uses the product of disjoint odd-valuation radicals. It does not replace an odd-valuation radical by an ordinary radical. At the removed primes, an arbitrary positive valuation contributes at most one prime factor to the comparison with M_5.

3 A compatible rational normalization

The following identities describe how the larger contact set can enter a local valuation construction. They are algebraic identities; they do not assert that an arbitrary circuit using them has the projection property of a defining polynomial.

Let $d = 5r$ and set $\epsilon = 1$ for odd r, $\epsilon = 2$ for even r. Thus $d + \epsilon$ is even and $1 \le \epsilon \le 2$. Put

$$F_r(t) = \prod_{\theta \in \Theta_r} (t - \theta).$$

Fix a positive rational square a, take $t = a\lambda$, and write $\delta = t - 1$. When $\delta F_r(t) \ne 0$, define

$$R = \frac{F_r(t)}{a^d \delta^{d+\epsilon}}, \qquad e = \frac{(a\delta)^d}{F_r(t)}, \qquad q = \frac{1}{R\delta^4}. \tag{4}$$

The corresponding cleared identities are

$$F_r(t)e = (a\delta)^d, \qquad q\delta^{4-\epsilon} = e, \qquad e\delta^{\epsilon} = R^{-1}. \tag{5}$$

Proposition 3.1. *The following assertions hold.*

1. *Every solution of the first identity in (5) has $F_r(t) \ne 0$. If $q = 0$ in the first two identities, then $\delta = 0$ and $M_r(t) = 1$.*
2. *If $p \mid M_r(t)$, then δ is a p-adic unit and $v_p(R)$ is odd. In particular, $v_p(q) = -v_p(R)$.*
3. *For $\lambda \to +\infty$,*
$$R \sim a^{-(d+\epsilon)} \lambda^{-\epsilon} > 0, \qquad q \sim a^{d+\epsilon-4} \lambda^{\epsilon-4} \longrightarrow 0.$$
4. *At a fixed prime where $v_p(t) < 0$ and is smaller than every valuation of a nonzero contact, one has*
$$v_p(q) = d\, v_p(a) - (4 - \epsilon) v_p(t).$$
For fixed a, this tends to $+\infty$ as $v_p(\lambda)$ tends to $-\infty$.

Proof. If $F_r(t) = 0$, then $t \in \Theta_r$, whence $\delta \neq 0$ and $(a\delta)^d \neq 0$, a contradiction. If $q = 0$, the second identity gives $e = 0$, and the first gives $\delta = 0$; Theorem 1.2 gives $M_r(t) = 1$.

At a consumed prime, exactly one contact difference has positive odd valuation, all other contact differences are units, and $t - 1$ is a unit. These facts follow from the determinant and $L_i(1, 1)$ conditions in Σ_r. Since $v_p(a)$ is even, the valuation $v_p(F_r(t)) - d\, v_p(a)$ is odd. This proves the second assertion. The third follows by comparing the leading terms in (4). For the fourth, every contact difference and δ has valuation $v_p(t)$. Thus $v_p(e) = d\, v_p(a)$, and the second identity in (5) proves the formula. □

Proposition 3.2. *Suppose $\lambda = u/v$ with v a positive integral square and $u \in \mathbb{Z}$, and suppose the d numbers*

$$p_\theta = u - \frac{\theta}{a} v \qquad (\theta \in \Theta_r)$$

are positive primes. Assume they avoid Σ_r and every prime dividing the numerator or denominator of a. If they are distinct, then R is positive for λ sufficiently large and its reduced denominator is a square. Its primes of odd valuation are exactly the p_θ, with valuation one. If in addition each p_θ is a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, and $\mathbb{Q}_{17}$, then so is R.

Proof. Direct substitution gives

$$R = \frac{\prod_{\theta \in \Theta_r} p_\theta}{v^d \delta^{d+\epsilon}}.$$

The denominator is a rational square. We check that it is a unit at each p_θ. Such a prime cannot divide v: otherwise it also divides u and hence all the distinct primes $p_{\theta'}$, since all contact and scale denominators are units there. Thus $a\lambda$ reduces to θ at this prime, and δ reduces to $\theta - 1 \neq 0$. Its denominator is therefore a unit, and the valuation of R is one. At every other prime the valuation is even. This proves the assertion about the reduced denominator and the odd valuations. Positivity follows from $d + \epsilon$ even and the positive numerator. The final assertion follows because a product of local squares divided by a nonzero square is a square. □

The prime-pattern hypothesis in Proposition 3.2 is the same finite-pattern requirement used in the arithmetic construction: the contacts and scales are fixed first, rational coefficients are cleared by a fixed square denominator, and a finite set of primes is avoided [2, 3]. Increasing r changes this finite list and the associated constants. An explicit numerical cutoff must consequently be recalculated.

4 The divisibility exponents required by height arguments

Lemma 4.1. *Let $\alpha > 0$, $H_* > 0$, and suppose a radical M satisfies $h(s) \leq H_* M(s)^\alpha$ at two nonzero rational values $s = \lambda$ and $s = T^2\lambda$, where $T \neq 0$. If $N \neq 0$ is an integer and*

$$M(\lambda)^k \mid N, \qquad M(T^2\lambda)^k \mid N,$$

then

$$h(T) \leq H_* |N|^{\alpha/k}.$$

The two radicals need not be relatively prime.

Proof. The elementary height inequalities give

$$2h(T) = h(T^2) \leq h(T^2\lambda) + h(\lambda).$$

Each radical is at most $|N|^{1/k}$, proving the result. □

Corollary 4.2. *Assume* (1). *Suppose* $B \geq 1$, $A > 0$, $J \geq 0$, $T, \lambda \in \mathbb{Q}^{\times}$, *and*

$$B^2 \leq Ah(T) + J.$$

Assume both λ *and* $T^2\lambda$ *avoid* Θ_r. *Let an integer residual* Δ *satisfy*

$$|\Delta| \leq 5B^4, \qquad M_r(\lambda)^k \mid \Delta, \qquad M_r(T^2\lambda)^k \mid \Delta.$$

If $rk > 38$, *then all solutions with* B *sufficiently large in terms of* r, k, A, J, H_r *satisfy* $\Delta = 0$.

Proof. If $\Delta \neq 0$, Theorem 1.2 and Lemma 4.1 imply

$$B^2 \leq AH_r 5^{19/(rk)} B^{76/(rk)} + J.$$

Since $76/(rk) < 2$, this bounds B above by a constant depending only on the displayed data. □

Corollary 4.3. *Under the same hypotheses, suppose an integer* $N \neq 0$ *instead satisfies*

$$|N| \leq 2\exp(h(\beta))B, \qquad M_r(\lambda)^{\mu} \mid N, \qquad M_r(T^2\lambda)^{\mu} \mid N.$$

If $2r\mu > 19$, *then*

$$\log B \leq \frac{19}{2r\mu - 19}\, h(\beta) + C,$$

where C *depends only on* r, μ, A, J, H_r.

Proof. Put $\gamma = 19/(r\mu) < 2$. The preceding argument gives

$$B^2 \leq AH_r 2^{\gamma} \exp(\gamma h(\beta)) B^{\gamma} + J.$$

If $B^2 \leq 2J$, the assertion follows by increasing C. Otherwise absorb J into $B^2/2$, take logarithms, and divide by $2 - \gamma$. The coefficient of $h(\beta)$ becomes $\gamma/(2-\gamma) = 19/(2r\mu - 19)$. □

The least positive integers meeting these two strict inequalities are

$$k_r = \left\lfloor \frac{38}{r} \right\rfloor + 1, \qquad \mu_r = \left\lfloor \frac{19}{2r} \right\rfloor + 1.$$

Some useful values are:

r	1	2	3	4	5	6	7	10	13	20	39
$5r$	5	10	15	20	25	30	35	50	65	100	195
k_r	39	20	13	10	8	7	6	4	3	2	1
μ_r	10	5	4	3	2	2	2	1	1	1	1

These are requirements on divisibility of integer residuals. A formal approximation formula must separately be proved accurate enough to produce those divisibilities. Lower exponents can shorten such a formula, but the enlarged product F_r requires additional arithmetic operations. Neither this table nor Theorem 1.2 is a coordinate count for a polynomial definition of the integers.

5 Exact finite verification

The accompanying standard-library program `check_contact_height_amplification.py` reconstructs the linear forms for the values of r in the table, verifies their primitivity and pairwise distinct zeros, and factors every integer defining Σ_r. It records the exceptional primes larger than 76176, the exact product E_r, and the height-transformation norm. It also checks the integer threshold inequalities and the normalized valuation identities. This calculation verifies the finite algebra; the arithmetic premise (1) remains explicit.

References

[1] E. Glazer and contributors, *Diophantine definition of the integers in the rationals*, `manuscripts/quartic_v9.tex`, commit `14e8f601fe0d9173d4a5f084aedc3e07066a7688`, https://github.com/elliotglazer/diophantine-z-q.

[2] Z.-W. Sun, *86 unknowns for Hilbert's tenth problem over the rationals*, arXiv:2610.20691v2, Theorem 4.1, https://arxiv.org/abs/2610.20691v2.

[3] J. Brox and contributors, *Arithmetic dependencies and the corrected source argument*, `arithmetic_dependencies_20261108_v8.tex`, Sections 2–3, https://github.com/JoseBrox/diophantine-z-q_improvements.